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Instruction

What the question is

Class P-Time

P membership: 7.8, 7.9, 7.10 -- ALLDFA is defined in problem 4.3

4.3 Let $ALL_{DFA} = \{\langle A \rangle \mid A \text{ is a DFA and } L(A) = \Sigma^*\}$. Show that ALL_{DFA} is decidable.

7.8 Let $CONNECTED = \{\langle G \rangle \mid G \text{ is a connected undirected graph}\}$. Analyze the algorithm given on page 185 to show that this language is in P.

7.8 On page 185 & Algorithm. A problem is said to be in P when method for solving the problem runs in time n^k where k is a constant or 187 ?

- In the 1st line, 1st node is selected & marked. This will require a constant time.
- In 2nd line & 3rd line, the nodes are marked until all the nodes are marked. If there are n nodes, then scanning of the nodes is there from the 1st 2, 3, 4, ... up to $n-1$. For every node, the order of neighbors will be order of n . This will result into the order of n^3 inspections.
- In line 4, all nodes are scanned for determining whether all the nodes are marked or not. This will take time = to order on n .

As the order is " $=$ " to n^3 , so this algorithm belongs to P .

7.9 A **triangle** in an undirected graph is a 3-clique. Show that $TRIANGLE \in P$, where $TRIANGLE = \{\langle G \rangle \mid G \text{ contains a triangle}\}$.

7.9] Class P: P is a class of Lang's that are decidable in polynomial time on a deterministic single-tape T.M.

Specified lang

• A triangle is an undirected graph is a 3-clique
 $\text{TRIANGLE} = \{ \langle G \rangle \mid G \text{ contains a triangle} \}$.

• Now we have to show that $\text{TRIANGLE} \in P$

• Let A be the T.M that decides TRIANGLE is polynomial time

Anyway, AS follows...

$A =$ On input $\langle G \rangle$, $\langle V, E \rangle$:

— Denotes set of _____ of the graph G .

V	Vertices
E	Edges

1. For $u, v, w \in V$ & $u < v < w$, we enumerate all triples $\langle u, v, w \rangle$.

2. Check whether all 3 edges $(u, v), (v, w) \& (w, u)$ exist in E or not. If exist then accept.

3. Otherwise reject.

• Enumeration of all triple require $O(|V|^3)$ time

• Checking whether all 3 edges belong to E take $O(|E|)$ time.

• Overall time is $O(|V|^3|E|)$ which is polynomial in the length of the input

Therefore $\text{TRIANGLE} \in P$

7.10 Show that ALL_{DFA} is in P.

7.10 Class P: P is a class of lang's that are decidable in Polynomial time on a deterministic single-tape Turing-machine.

$ALL_{DFA} = \{ \langle A \rangle \mid A \text{ is a DFA that recognizes } \Sigma^* \}$

$E_{DFA} = \{ \langle A \rangle \mid A \text{ is a DFA } \& L(A) = \emptyset \}$ is determined by Let E be the T.M that determines E_{DFA} .

Let R be the T.M that determines ALL_{DFA} . The algorithm of R is this...

1. On input $\langle A \rangle$, where A is a DFA, construct a DFA B that recognizes $\overline{L(A)}$, by swapping accept & non-accepting states.
2. Run the T.M E on input $\langle B \rangle$, where E determines E_{DFA} .
3. If E accepts, then accept.
4. If E rejects, then reject.

Clearly the T.M, R determines ALL_{DFA} in Polynomial time.

Therefore, ALL_{DFA} is in P.

7.6 Show that P is closed under union, concatenation, and complement.

7.6 The class P is closed under union, concatenation, & complement.
Class P : P is the class of Lang's that are decidable in polynomial time on a deterministic single-tape T.M. That = $P = \text{TIME}(n^k)$
Now we have to show that P is closed under union, concatenation, & complement.

① Union

Assume 2 Lang's $P_1 \in P$ & $P_2 \in P$. The T.M. M that accepts $P_1 \cup P_2$ works as follows:

$M =$ "on input w ;

1. Check if $w \in P_1$
2. If not then check if $w \in P_2$
3. Accept w if & only if P_1 or P_2 accepts
4. If both reject then reject the input w ."

Since each membership check requires polynomial time the overall time is polynomial.

② Concatenation

Assume 2 Lang's $P_1 \in P$ & $P_2 \in P$
The T.M. M that accepts $P_1 P_2$ works as follows
 $M =$ "on input w of length n ;

1. w can be split into 2 strings in n different ways
 2. For each split
 - A) Check if the 1st substr. belongs to P_1
 - B) Check if 2nd substr. belongs to P_2
 3. If any split succeeds, then accept."
- Clearly the overall time is polynomial

7.6 ③ complement

Assume a Lang $P_1 \in P$

The T.M M that accepts $\bar{P}_1$ works as follows
 $M \Rightarrow$ on input w

1. Check if $w \in P_1$,
2. If $w \in P_1$, then reject
3. If $w \notin P_1$, then accept

Clearly the overall time is polynomial
therefore, the class P is closed under union
concatenation, & complement.